Confluent Cloud Terraform Deployment — Timer-Driven PTF UDFs (early access example)

> This example deploys the **Session Timeout Detector**, **Abandoned Cart Detector**, **Per-Event Follow-Up**, and **SLA Monitor** PTF UDFs (source in examples/ptf_udf_timer_driven/java/) to **Confluent Cloud** using Terraform. All four UDFs are packaged in a single JAR. All infrastructure (environment, Kafka cluster, Flink compute pool, service accounts, API keys) and Flink SQL statements are declared as Terraform resources.

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1.0 Overview

1.1 How this differs from the other deployment paths

Aspect	Confluent Cloud Terraform (this example)	Confluent Platform SQL Client
Where it runs	Confluent Cloud	Confluent Platform on Minikube
How SQL is submitted	`confluent_flink_statement` Terraform resources	`sql-client.sh -f` on the JobManager pod
UDF JAR delivery	`confluent_flink_artifact` (uploaded to CC)	`kubectl exec` to Flink pods
Entry point	`make deploy-cc-ptf-udf-timer-driven`	`make deploy-cp-ptf-udf-timer-driven`
Requires code compilation	✅ (Java + Gradle for UDF JAR)	✅ (Java + Gradle for UDF JAR)

Aspect	Confluent Cloud Terraform (this example)	Confluent Platform SQL Client
Statement lifecycle	Managed by Terraform state	Managed by Flink session cluster
Same codebase for both?	✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment)	✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment)

2.0 How it works

2.1 Infrastructure provisioning

Terraform declares the full Confluent Cloud stack:

Resource	Purpose
`confluent_environment`	Isolated CC environment (`ptf-udf-timer-driven`)
`confluent_kafka_cluster`	Standard single-zone Kafka cluster (AWS us-east-1)
`confluent_schema_registry_cluster`	Stream Governance Essentials (auto-provisioned with environment)
`confluent_service_account`	`ptf-udf-timer-driven` service account with FlinkDeveloper, ResourceOwner, and Assigner roles
`confluent_flink_compute_pool`	Flink compute pool (max 10 CFU)
`confluent_api_key`	Flink-specific API key for statement submission
`flink_api_key_rotation` module	Rotating API key pair for the Flink service account

2.2 JAR delivery

On Confluent Cloud, UDF JARs are uploaded as **Flink artifacts** via the `confluent_flink_artifact` resource. The JAR is built from examples/ptf_udf_timer_driven/java/ and uploaded directly from the local build output. All four `CREATE FUNCTION ... USING JAR` statements reference the same artifact using a `confluent-artifact://` URI, registering `session_timeout_detector`, `abandoned_cart_detector`, `per_event_follow_up`, and `sla_monitor` from the single JAR.

2.3 Statement flow

Terraform manages the SQL statements as `confluent_flink_statement` resources with explicit `depends_on` ordering:

```
┌──────────────────────────────────────────────────────────────────────┐
│  ── Named timer pipeline (SessionTimeoutDetector) ──                 │
│  Step 1:  DROP TABLE IF EXISTS user_activity            → OK         │
│  Step 2:  CREATE TABLE user_activity (...)              → OK         │
```

```
│  Step 3:  INSERT INTO user_activity VALUES (sample data) → submitted │
│  Step 4:  DROP TABLE IF EXISTS timeout_events           → OK        │
│  Step 5:  CREATE TABLE timeout_events (...)             → OK        │
│                                                                     │
│  ── Unnamed timer pipeline (PerEventFollowUp) ──                    │
│  Step 6:  DROP TABLE IF EXISTS user_actions             → OK        │
│  Step 7:  CREATE TABLE user_actions (...)               → OK        │
│  Step 8:  INSERT INTO user_actions VALUES (sample data) → submitted │
│  Step 9:  DROP TABLE IF EXISTS follow_up_events         → OK        │
│  Step 10: CREATE TABLE follow_up_events (...)           → OK        │
│                                                                     │
│  ── SLA monitoring pipeline (SlaMonitor) ──                         │
│  Step 6:  DROP TABLE IF EXISTS service_requests         → OK        │
│  Step 7:  CREATE TABLE service_requests (...)           → OK        │
│  Step 8:  INSERT INTO service_requests VALUES (...)     → submitted │
│  Step 9:  DROP TABLE IF EXISTS sla_events               → OK        │
│  Step 10: CREATE TABLE sla_events (...)                 → OK        │
│                                                                     │
│  ── Shared JAR upload and function registration ──                  │
│  Step 11: Upload UDF JAR as confluent_flink_artifact    → OK        │
│  Step 12: CREATE FUNCTION session_timeout_detector      → OK        │
│  Step 13: INSERT INTO timeout_events                    → submitted │
│  Step 14: CREATE FUNCTION per_event_follow_up           → OK        │
│  Step 15: INSERT INTO follow_up_events                  → submitted │
│  Step 16: CREATE FUNCTION sla_monitor                   → OK        │
│  Step 17: INSERT INTO sla_events                        → submitted │
│                                                                     │
│  ── Abandoned cart pipeline (AbandonedCartDetector) ──              │
│  Step 18: DROP TABLE IF EXISTS cart_events              → OK        │
│  Step 19: CREATE TABLE cart_events (...)                → OK        │
│  Step 20: INSERT INTO cart_events VALUES (sample data)  → submitted │
│  Step 21: DROP TABLE IF EXISTS abandoned_cart_events    → OK        │
│  Step 22: CREATE TABLE abandoned_cart_events (...)      → OK        │
│  Step 23: CREATE FUNCTION abandoned_cart_detector       → OK        │
│  Step 24: INSERT INTO abandoned_cart_events             → submitted │
└─────────────────────────────────────────────────────────────────────┘
```

Steps 13, 15, 17, and 24 are **long-running streaming jobs**. They run continuously, reading from their respective source topics and writing output to sink topics.

Once deployment completes, Terraform generates a visual **resource graph** at `examples/ptf_udf_timer_driven/cc_deploy/terraform.png`, providing an at-a-glance view of the infrastructure and resource dependencies:

3.0 Prerequisites

- macOS with Homebrew or Linux with apt-get

- Java 21 and Gradle installed (for building the UDF JAR)
- Terraform installed
- A Confluent Cloud account with a **Cloud API key** and **secret** (create one here)
- The UDF JAR must be built before deploying:

```
make build-ptf-udf-timer-driven     # builds the UDF fat JAR from
examples/ptf_udf_timer_driven/java/
```

4.0 How to run

All commands are run from the **project root** (where the `Makefile` lives).

4.1 Deploy

A single target builds the UDF JAR and runs `terraform apply`:

```
make deploy-cc-ptf-udf-timer-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

Behind the scenes this runs:

Step	What it does
1	`./gradlew clean shadowJar` — builds the UDF fat JAR from `examples/ptf_udf_timer_driven/java/`
2	`terraform init` — initializes the Terraform working directory
3	`terraform apply -auto-approve` — provisions all CC infrastructure and submits Flink SQL statements
4	Generates a Terraform visualization at `examples/ptf_udf_timer_driven/cc_deploy/terraform.png`

4.2 Monitor

Monitor the running Flink statements in the Confluent Cloud Console:

1. Navigate to your **ptf-udf-timer-driven** environment
2. Open the **Flink** tab to see compute pools and running statements
3. Open the **Topics** tab to inspect `user_activity`, `timeout_events`, `user_actions`, `follow_up_events`, `service_requests`, `sla_events`, `cart_events`, and `abandoned_cart_events`

4.3 Tear down

To destroy all Confluent Cloud resources created by Terraform:

```
make teardown-cc-ptf-udf-timer-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

This runs `terraform destroy -auto-approve`, removing all Flink statements, the compute pool, Kafka cluster, service accounts, and the environment.

5.0 Troubleshooting

If you encounter an error during `make deploy-cc-ptf-udf-timer-driven` that looks like the following, it is because your organization does not have early access to timer services for PTF UDFs. Please contact your Confluent Account Executive to request access:

```
│ Error: error waiting for Flink Statement "tf-2026-04-02-100522-54a7b07c-
4a42-4d53-90e0-034cb2ae1ea4" to provision: Flink Statement "tf-2026-04-02-
100522-54a7b07c-4a42-4d53-90e0-034cb2ae1ea4" provisioning status is
"FAILED": Method 'onTimer' of function class 'ptf.SessionTimeoutDetector'
- Timer service is not supported for PTFs.; could not parse error details;
raw response body: "{\"api_version\":\"sql/v1\",\"environment_id\":\"env-
m60dnq\",\"kind\":\"Statement\",\"metadata\":{\"created_at\":\"2026-04-
02T14:05:22.440172Z\",\"labels\":
{},\"resource_version\":\"5\",\"self\":\"https://flink.us-east-
1.aws.confluent.cloud/sql/v1/organizations/bd545cc3-8e7c-4387-b6d8-
b6d1497a9df7/environments/env-m60dnq/statements/tf-2026-04-02-100522-
54a7b07c-4a42-4d53-90e0-034cb2ae1ea4\",\"uid\":\"eff86f2f-bb53-474d-87bd-
3300ead6db9f\",\"updated_at\":\"2026-04-
02T14:05:22.64651Z\"},\"name\":\"tf-2026-04-02-100522-54a7b07c-4a42-4d53-
90e0-034cb2ae1ea4\",\"organization_id\":\"bd545cc3-8e7c-4387-b6d8-
b6d1497a9df7\",\"spec\":{\"compute_pool_id\":\"lfcp-
nnpz1v\",\"execution_mode\":\"STREAMING\",\"principal\":\"sa-
nvj798k\",\"properties\":{\"sql.current-catalog\":\"ptf-udf-timer-
driven\",\"sql.current-database\":\"ptf-udf-timer-
driven\"},\"statement\":\"CREATE FUNCTION IF NOT EXISTS
session_timeout_detector\\n  AS 'ptf.SessionTimeoutDetector'\\n  USING JAR
'confluent-artifact://cfa-q57pk2';\\n\",\"stopped\":false},\"status\":
{\"detail\":\"\",\"network_kind\":\"PUBLIC\",\"phase\":\"PENDING\",\"trait
s\":{\"connection_refs\":
[],\"is_append_only\":true,\"is_bounded\":true,\"schema\":
{},\"sql_kind\":\"CREATE_FUNCTION\",\"upsert_columns\":null},\"warnings\":
[]}}"
│
│   with confluent_flink_statement.create_session_timeout_detector,
│   on setup-confluent-flink.tf line 340, in resource
"confluent_flink_statement" "create_session_timeout_detector":
│  340: resource "confluent_flink_statement"
"create_session_timeout_detector" {
```

6.0 Resources

- Confluent Terraform Provider
- confluent_flink_statement Resource
- confluent_flink_artifact Resource
- Process Table Functions (PTFs)
- Create a User-Defined Function with Confluent Cloud for Apache Flink